

DAYANANDA SAGAR UNIVERSITY, BENGALURU

School of Engineering • End Semester Examination

Course: Applied Cryptography and Network Security | Course Code: 21CY51 | Credits: 4

Exam Session: End Semester Exam 2024

Max Marks: 80 | Duration: 3 Hours

INSTRUCTIONS TO CANDIDATES:

1. Answer FIVE full questions, choosing ONE full question from each module.
2. Draw neat diagrams wherever necessary. Missing data may be suitably assumed.

MODULE 1

Q1. In RSA cryptosystem, select primes $p = 7$ and $q = 11$. Choose encryption key $e = 13$. (i) Calculate public key and private key d , (ii) Encrypt plaintext $M = 9$, and (iii) Decrypt ciphertext C to retrieve original message.

[8 Marks]